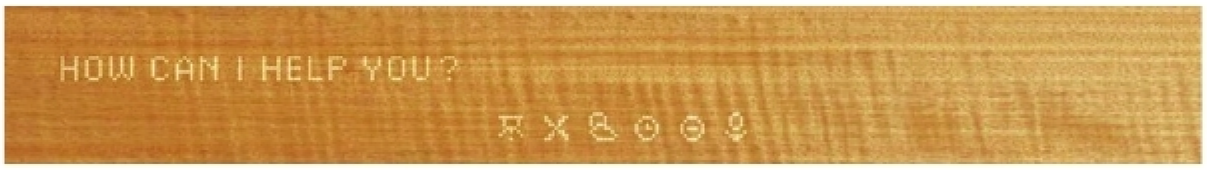

Figure 1: mui smart wooden plank (Credit: www.kickstarter.com)

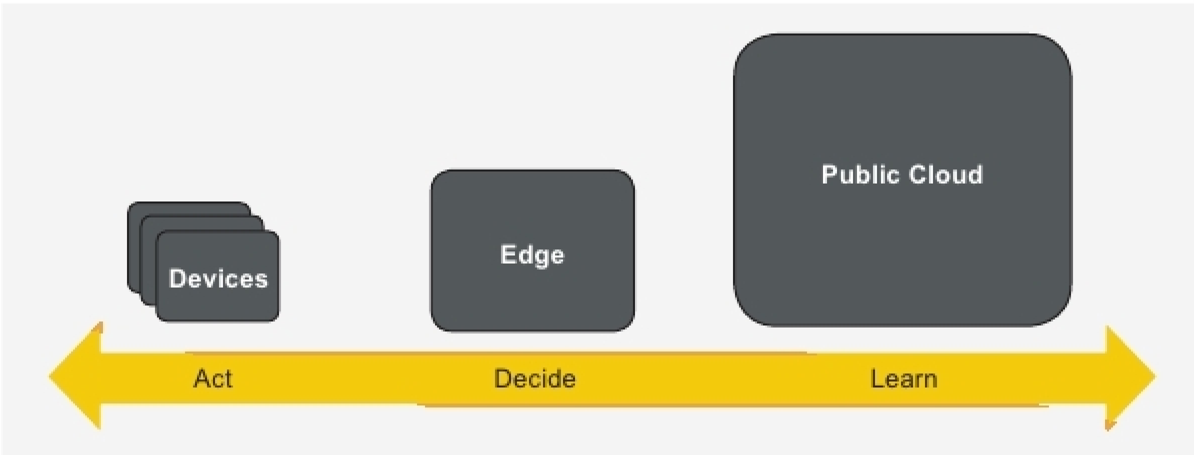

Figure 2: Edge computing architecture (Credit: thenewstack.io)

manufacturing. With the explosion of IoT devices, unprecedented amounts of data generated by the IoT daily across the globe will have a huge impact across the entire Big Data universe.

Since data volume and complexity are growing everyday, the IoT Big Data has thrown new security challenges. One cannot easily distinguish between useful, harmful and redundant data. Collecting data is an easy process as compared to analysing it. To use data in a meaningful way, it must first be checked and processed with advanced tools (analytics and algorithms). This is where the role of AI comes in.

AI often revolves around the use of simple to complex algorithms. Heaps of data can be managed and used efficiently and effectively using AI. Many vendors of IoT platform software are now offering integrated AI capabilities. IoT devices with AI could be a good combination to serve humankind in a better way.

AIOps—a new term coined by Gartner—is the outcome of disruptive technologies. AIOps platforms utilise Big Data, machine learning and other advanced analytics technologies to support and enhance IT operation processes, including monitoring, analysis, automation and service management. These platforms enable concurrent use of multiple data sources, data-collection methods, and analytical (real-time and deep) and presentation technologies.

Big Data. Big Data is the name given to huge volumes of data that can be analysed to get useful patterns and make business predictions. The IoT Big Data is found in many applications including predictive models, analysis, smart devices and machine learning.

AI is required to make sense of Big Data that is generated from various sensors. Hence, it becomes a prerequisite for an IoT system to work intelligently for analysing, collecting, manipulating and generating insights, from the enormous amount of data at high speed.

Edge computing. Edge computing is important in machine learning. It makes decisions by applying intelligence based on the deployed machine learning models. For example, IoT devices are deployed for monitoring gas pipelines. Here, costly human inspection is replaced by technology that can rapidly detect leakages or other anomalies, and automatically alert the concerned person(s) for appropriate action. In such cases, distributed computing at the edge can dramatically cut response times.

Edge computing architecture can be visualised using a three-tier architecture (Figure 2). The first layer has local devices and applications, the second is the edge layer and the third is public cloud. Devices are sensors and actuators that are responsible for collecting data and controlling devices. These are connected to the cloud through the local edge computing layer.

Many companies are investing in the convergence of the IoT and AI. For example, Microsoft is working on its vision for Intelligent Edge. Its Azure IoT Edge platform enables low-power devices to run containers and perform AI locally while retaining a connection to the cloud for management and modelling. Azure IoT Edge is an open source project available on GitHub.

Amazon Web Services (AWS) launched its edge computing platform, Greengrass, to incorporate machine learning. It is built to deliver capabilities similar to Azure IoT Edge.

Smart homes. Smart devices like phones, computers, home appliances and the like are embedded with sensors, software and actuators. Connectivity and communication between devices are possible using the IoT. Through the combination of AI and the IoT, one can instruct smart assistants like Alexa to control devices.

Another example is a smart thermostat connected to the Internet. Some smart thermostats include intelligence that can learn when the house is likely to be occupied. This allows automatic pre-heating, so that room temperature is comfortable when a resident arrives. If lifestyle of the resident changes, the thermostats can gradually adjust the schedule, maintaining energy savings. A combination of AI and the IoT can teach decision-making to machines. These machines can act as smart assistants, like Alexa, to control smart home devices.

Smart security cameras. These are self-contained, standalone vision systems with a built-in image sensor. These utilise the Internet and intelligent programs that analyse images from cameras to recognise humans, vehicles or objects. One can do real-time monitoring remotely from a phone or computer. Latest automated surveillance with new machine learning technique could give CCTV cameras the ability to spot troublesome and abnormal behaviour without human supervision.